



SITEFLOW AI

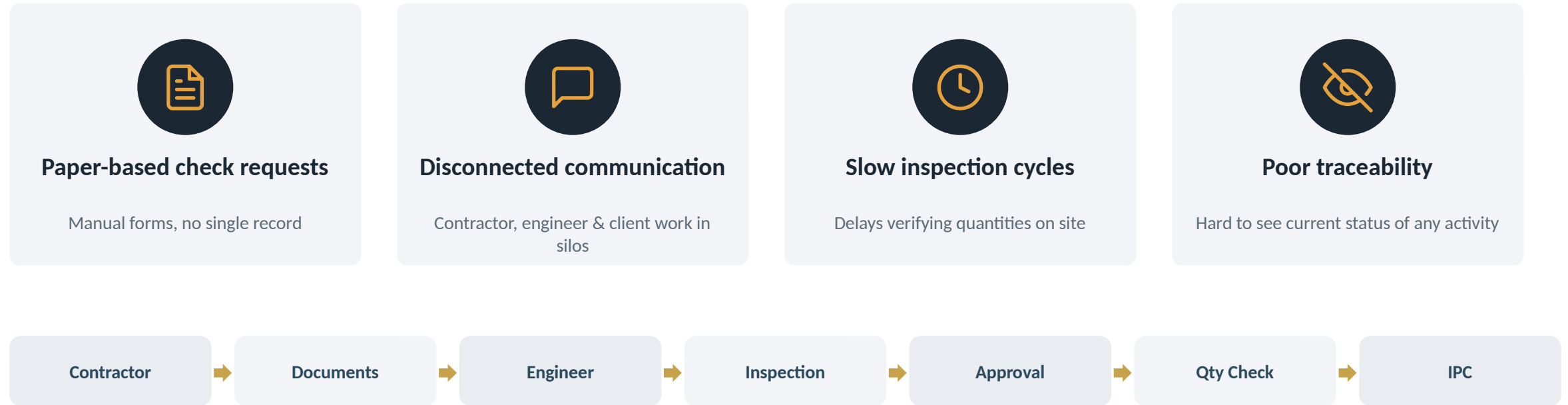
AI-Powered Construction Inspection, Approval & IPC Management

From Check Request to Certified Payment.

Construction Workflow Digitization • Hackathon MVP • 6-Member Team, 48 Hours

github.com/SidraPervaiz1122/SiteFlowAI

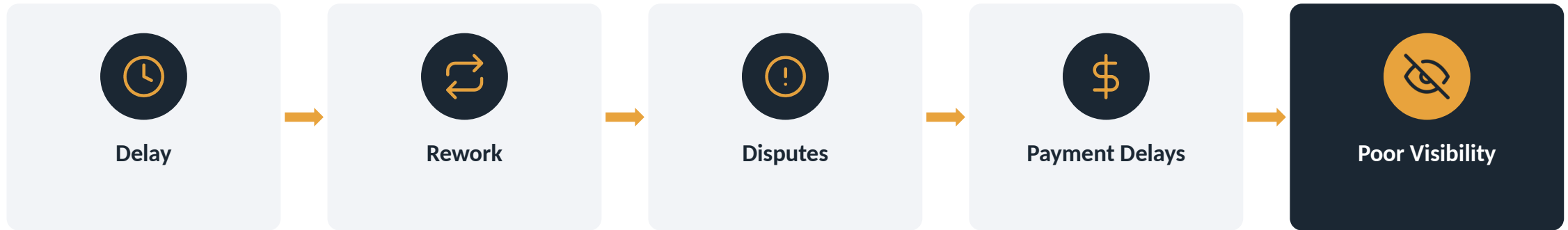
Construction workflows are fragmented by design



Each handoff above is manual, undocumented, and easy to lose — the real source of delay.

"The inspection process, quantity verification process, and payment process are disconnected." — SiteFlow AI Product Requirements Document

Fragmentation has a compounding cost

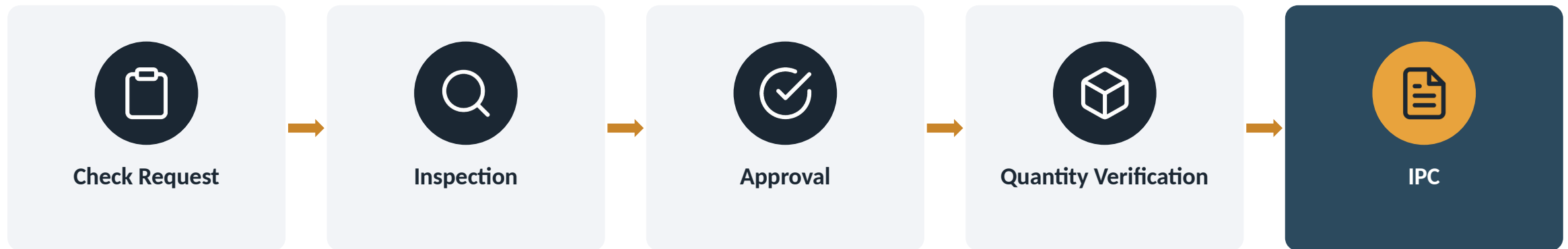


Each unresolved step upstream doesn't disappear — it resurfaces later as rework, a dispute over quantities, or a stalled Interim Payment Certificate.

Documented consequences (SiteFlow AI PRD): duplication of effort, delays in quantity verification, delays preparing payment information, and poor activity traceability across the inspection-to-payment lifecycle.

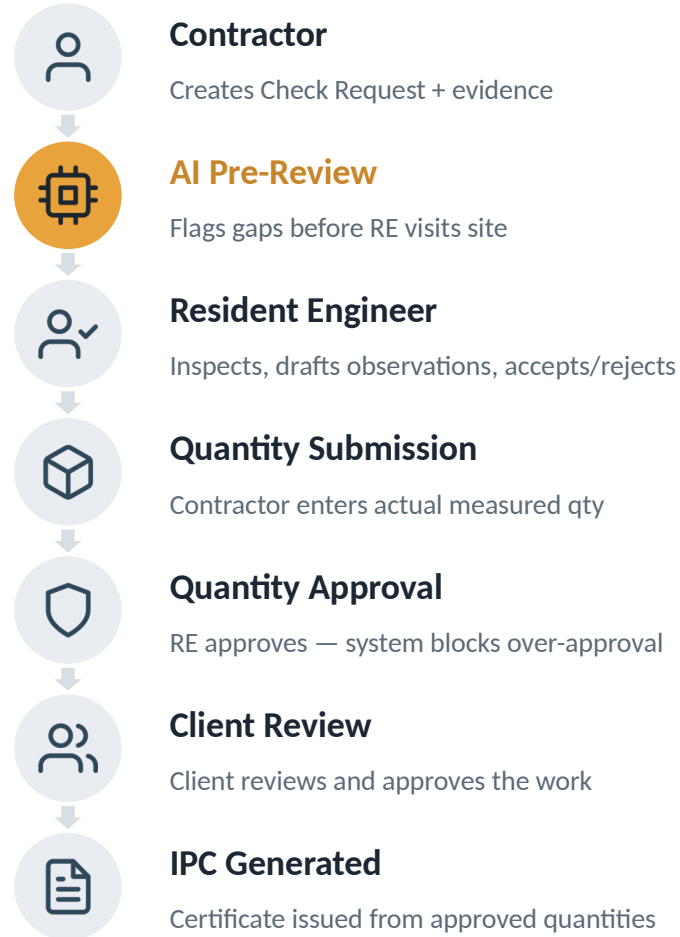
One connected digital workflow

SiteFlow AI links Contractor, Resident Engineer, and Client through a single structured pipeline — every check request, inspection, quantity, and payment step is recorded once and stays traceable end to end.



The Bill of Quantities is the fixed contractual baseline — quantities and rates never change mid-workflow. Every approval is checked against it automatically.

From site activity to payment — one traceable path



Where AI Enters

AI Pre-Review

Flags missing evidence and risk before the engineer visits.

Observation Drafting

Assists the RE in writing structured site observations.

Every AI output is advisory — a human always approves.

Three concrete AI capabilities — advisory only



Check Request Pre-Review

INPUT

Contractor's check request + attached evidence

AI PROCESS

AI evaluates completeness, technical risk, missing test reports/photos

OUTPUT

Readiness score, missing-evidence list, suggested inspection checks

VALUE

Engineer arrives prepared — fewer wasted site visits



Observation Drafting

INPUT

Site notes / defect hints + inspection photos

AI PROCESS

AI drafts a structured observation (title, severity, category)

OUTPUT

Editable draft observation for the Resident Engineer

VALUE

Faster, more consistent inspection records



Project Q&A (RAG)

INPUT

Natural-language question about the project

AI PROCESS

Retrieval-augmented answer grounded strictly in project documents

OUTPUT

Concise answer, or an explicit "not found" if unsupported

VALUE

Instant answers without hunting through documents

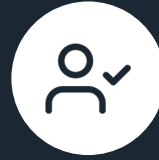
Guardrailed by design: AI can never approve, reject, or modify quantities, rates, or IPC payments — that authority stays with the Resident Engineer and Client.

Three roles, one shared source of truth



Contractor

- Submits Check Requests
- Attaches evidence & photos
- Enters actual measured quantities
- Tracks inspection status



Resident Engineer

- Reviews & inspects on site
- Drafts observations (AI-assisted)
- Accepts / rejects requests
- Approves verified quantities



Client

- Reviews approved work
- Approves for payment
- Monitors overall progress
- Receives generated IPC

A layered, production-shaped stack



Frontend

React 18 · TypeScript 5 · Vite 5 · React Router 6

Role-based UI, Quick Demo Role Switcher



Backend / API

FastAPI · Pydantic 2 · PyJWT · Passlib (bcrypt)

REST API, JWT auth, RBAC enforcement



AI / Intelligence Layer

Guardrailed prompt templates · Hybrid RAG retriever

Pre-review, observation drafting, project Q&A



Database

SQLAlchemy 2 ORM · SQLite (dev)

Atomic, transaction-safe quantity validation



Deployment

Docker (multi-stage build)

Azure Container Registry

Azure App Service

GitHub Actions CI/CD:

build → push → restart → automated health
check on every merge to main.

One activity, followed from site to certificate

BOQ Item #8 – Brick / Block Masonry · Contract Qty: 30.79 m³ · Rate: Rs. 15,500/m³ · Contract Value: Rs. 477,245

-  1 Contractor completes masonry work and submits a Check Request with site photos.
-  2 AI pre-review flags evidence completeness and highlights inspection focus points.
-  3 Resident Engineer inspects on site and drafts an observation (AI-assisted).
-  4 Engineer accepts the request; contractor submits actual measured quantity.
-  5 System validates the quantity against remaining BOQ balance — atomically.
-  6 Client reviews and approves the verified, quantity-backed work.
-  7 Approved quantity flows into IPC generation — payment-ready, fully traceable.

What's actually built



Check Request Management

Create, evidence upload, submission tracking



Inspection Workflow

On-site inspection with structured observations



Approval Workflow

Accept / reject / resubmit with full history



Atomic Quantity Validation

DB-level over-approval prevention



IPC Generation

Certificates from authoritative approved qty



AI Assistance

Pre-review, observation drafting, RAG Q&A



Role-Based Access Control

JWT-secured Contractor / RE / Client roles



Evidence & Document Management

File/photo attachments per check request



Audit Trail

Every key mutation generates an audit event

What makes SiteFlow AI different



The stack behind the workflow



Frontend

- React 18.2
- TypeScript 5.2
- Vite 5.1
- React Router 6.22
- Lucide Icons



Backend

- FastAPI 0.110+
- Pydantic 2.6+
- SQLAlchemy 2.0+
- PyJWT + Passlib/bcrypt



AI Layer

- Guardrailed prompts
- Hybrid RAG retriever
- Forbidden-phrase filter



Infrastructure

- Docker (multi-stage build)
- Azure Container Registry
- Azure App Service
- GitHub Actions CI/CD

What this enables — today and next



Today — Implemented Value

- Faster inspections — pre-review flags gaps before the site visit
- Reduced administrative overhead — one workflow, not three records
- Better transparency — every stakeholder sees the same live status
- Fewer disputes — quantities are validated atomically against contract data
- Faster IPC preparation — certificates generated directly from approved data



Next — Future Work

- *Predictive project risk analysis*
- *Computer vision for automated site inspection*
- *Automated quantity estimation from photos*
- *Mobile field inspection app*
- *BIM & ERP integration*
- *Advanced project analytics dashboards*



*Construction shouldn't lose time to paperwork
and disconnected workflows.*

SITEFLOW AI

Digitizing the journey from site inspection to certified payment.

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Thank you.